

ZIHAO WANG

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INTERESTS

I work on building open-ended embodied agents with multi-task skills. In particular, I am interested in building and leveraging large pre-trained Agent Models to improve the generalization of agent capabilities. My long-horizon goal is to build interactive open-world generalist agents capable of understanding human instructions and executing tasks with human-like thinking and computer control.

Recently, we have developed a series of open-world multi-task agents, including **OmniJARVIS**, **JARVIS-VLA** (pretrained end-to-end Vision-Language-Action models), **JARVIS-1** (self-improving agent with multimodal memory), **DEPS** (interactive long-horizon planning agent), **RAT** (reasoning agent with retrieval-augmented thought), and **ProAgent** (collaborating agents).

EDUCATION

Peking University

Beijing, China

[Ph.D.](#) Student in Artificial Intelligence (AI)

2022.09 - Present

- Advisor: Professor Yitao Liang

Beijing Institute of Technology

Beijing, China

[M.S.](#) in Control Science and Technology

2019 - 2022

- Advisor: Professor Zhen Li

Beijing Institute of Technology

Beijing, China

[B.S.](#) in Automation

2015 - 2019

PROFESSIONAL EXPERIENCE

Research Intern, AI Lab in Ailibaba Inc, Beijing, CHINA

2021.06 - 2021.08

PUBLICATIONS

• Pre-Trained Agent Models

[1] [Zihao Wang](#), Shaofei Cai, Zhancun Mu, Haowei Lin, Ceyao Zhang, Xuejie Liu, Qing Li, Anji Liu, Xiaojian Ma, Yitao Liang. OmniJARVIS: Unified Vision-Language-Action Tokenization Enables Open-World Instruction Following Agents. NeurIPS 2024. [\[Page\]](#)

[2] [Muyao Li*](#), [Zihao Wang*](#), Kaichen He, Xiaojian Ma and Yitao Liang. JARVIS-VLA: Post-Training Large-Scale Vision Language Models to Play Visual Games with Keyboards and Mouse. *arXiv preprint arXiv:2311.05997*, 2023 [\[Page\]](#)

• LLM-Based Agent: Planning, Memory and Reasoning

[3] [Zihao Wang](#), Shaofei Cai, Anji Liu, Yonggang Jin, Jinbing Hou, Bowei Zhang, Haowei Lin, Zhaofeng He, Zilong Zheng, Yaodong Yang, Xiaojian Ma, Yitao Liang. JARVIS-1: Open-world multi-task agents with memory-augmented multimodal language models. *T-PAMI*, 2024. [\[Page\]](#)

[4] [Zihao Wang](#), Shaofei Cai, Guanzhou Chen, Anji Liu, Xiaojian Ma, Yitao Liang. Describe, Explain, Plan and Select: Interactive Planning with Large Language Models Enables Open-World

Multi-Task Agents.

NeurIPS, 2023; also appeared on **ICML 2023 TEACH Workshop Best Paper**. [\[Code\]](#)

[5] Zihao Wang, Anji Liu, Haowei Lin, Jiaqi Li, Xiaojian Ma, Yitao Liang. Retrieval Augmented Thoughts Elicit Context-Aware Reasoning in Long-Horizon Generation.

Open-world Agents Workshop in NeurIPS 2024. [\[Page\]](#)

[6] Ceyao Zhang, Kaijie Yang, Siyi Hu, Zihao Wang, Guanghe Li, Yihang Sun, Cheng Zhang, Zhaowei Zhang, Anji Liu, Song-Chun Zhu, Xiaojun Chang, Junge Zhang, Feng Yin, Yitao Liang, Yaodong Yang. ProAgent: Building Proactive Cooperative Agents with Large Language Models. AAAI 2024 (**Oral**). [\[Page\]](#)

- **Imitation Learning Policy in Hierarchical Agents**

[7] Shaofei Cai, Zihao Wang, Kewei Lian, Zhancun Mu, Xiaojian Ma, Anji Liu, Yitao Liang. ROCKET-1: Master Open-World Interaction with Visual-Temporal Context Prompting. CVPR 2025. [\[Page\]](#)

[8] Shaofei Cai, Bowei Zhang, Zihao Wang, Xiaojian Ma, Anji Liu, Yitao Liang. GROOT-2: Weakly Supervised Multi-Modal Instruction Following Agents. ICLR 2025. [\[Paper\]](#)

[9] Shaofei Cai, Bowei Zhang, Zihao Wang, Xiaojian Ma, Anji Liu, Yitao Liang. GROOT: Learning to Follow Instructions by Watching Gameplay Videos. ICLR 2024 (**Spotlight**). [\[Page\]](#)

[10] Shaofei Cai, Zihao Wang, Xiaojian Ma, Anji Liu, Yitao Liang. Open-world multi-task control through goal-aware representation learning and adaptive horizon prediction. CVPR, 2023. [\[Code\]](#)

- **Agent Evaluations**

[11] Haowei Lin, Zihao Wang, Jianzhu Ma, Yitao Liang. MCU: A Task-centric Framework for Open-ended Agent Evaluation in Minecraft. ICML, 2025 (**Spotlight**).

[12] Haowei Lin, Xiangyu Wang, Ruilin Yan, Baizhou Huang, Haotian Ye, Jianhua Zhu, Zihao Wang, James Zou, Jianzhu Ma, Yitao Liang. Generative evaluation of complex reasoning in large language models. arXiv: 2504.02810.

- **Visual Geometry and SLAM**

[13] Zihao Wang, Chunxu Wu, Yifei Yang, Zhen Li. Learning Transformation-Predictive Representations for Detection and Description of Local Features. CVPR, 2023.

[14] Zihao Wang, Zhen Li, Xueyi Li, Wenjie Chen, Xiangdong Liu. Graph-Based Contrastive Learning for Description and Detection of Local Features. IEEE Trans. Neural Netw. Learn. Syst. (TNNLS) 2022.

[15] Zihao Wang, Xueyi Li, Zhen Li Local Representation is NOT Enough: Soft Point-wise Transformer for descriptor and Detector of Local Features. IJCAI 2021.

- **Others**

[16] Haowei Lin, Baizhou Huang, Haotian Ye, Qinyu Chen, Zihao Wang, Sujian Li, Jianzhu Ma, Xiaojun Wan, James Zou, Yitao Liang. Selecting Large Language Model to Fine-tune via Rectified Scaling Law. ICML 2024.

[17] Yuheng Cheng, Ceyao Zhang, Zhengwen Zhang, Xiangrui Meng, Sirui Hong, Wenhao Li, Zihao Wang, Zekai Wang, Feng Yin, Junhua Zhao, Xiuqiang He. Exploring large language model based intelligent agents: Definitions, methods, and prospects. *arXiv:2401.03428*, 2024.

- [18] Haotian Zhang, Juntao Zhou, Haowei Lin, Hang Ye, Jianhua Zhu, Zihao Wang, Liangcai Gao, Yizhou Wang, Yitao Liang. CLoG: Benchmarking Continual Learning of Image Generation Models. *arXiv preprint arXiv:2406.04584*, 2024. [\[Code\]](#)
- [19] Jingwen Deng*, Zihao Wang*, Shaofei Cai, Anji Liu, Yitao Liang. Open-World Skill Discovery from Unsegmented Demonstrations. *arXiv preprint arXiv:2503.10684*, 2025.
- [20] Guangyu Zhao, Kewei Lian, Haowei Lin, Haobo Fu, Qiang Fu, Shaofei Cai, Zihao Wang, Yitao Liang. Optimizing Latent Goal by Learning from Trajectory Preference. *arXiv preprint arXiv:2412.02125*, 2024.

AWARDS

- **National Scholarship**, CHINA, 2021
- **Outstanding Graduate** (top 1%), Beijing, CHINA, 2019
- **Ranking 2nd** on ECCV 2024 Multimodal Perception and Comprehension of Corner Cases in Autonomous Driving Challenges.
- **Special Prize** on Autonomy in International Micro-unmanned Aerial Vehicle Competition (IMAV), Melbourne, Australia, 2018